

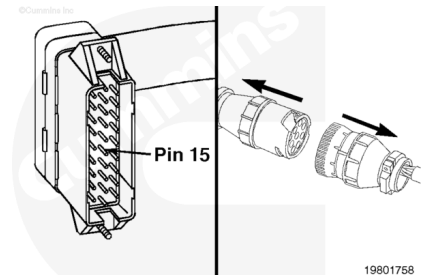
Trainings ▾

General Information

The portion of the alternate torque/alternate low-idle circuit in the engine harness consists of the signal wire connected to pin 15 of the main engine harness connector.

To check the OEM portion of the alternate torque/alternate low-idle switch circuit, refer to Procedure 019-071.

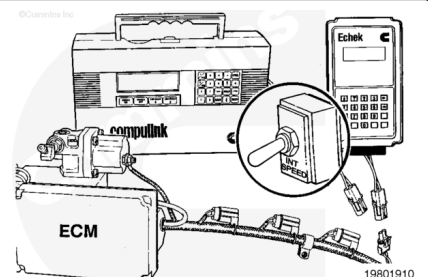
The switch is located on the driver interface panel.



Resistance Check

If an electronic service tool is available, then the switch should be monitored for proper operation. If the switch is changing state correctly on the service tool, then the problem does **not** lie in the switch circuit. If an electronic service tool is **not** available, check the switch manually.

Note : Conditions that activate this switch will vary between OEMs and applications. Some applications use a relay instead of a switch; some switches are mounted remotely. Refer to the OEM manual to see how a particular system is wired.

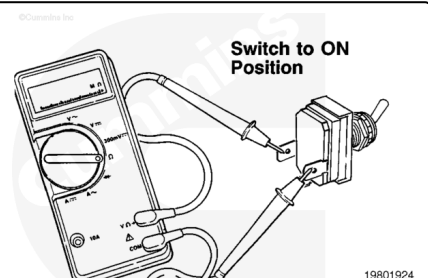


Locate the switch on the driver interface panel and remove it. Refer to the OEM troubleshooting and repair manual for the procedure.

Disconnect the wires connected to the switch (wire Nos. C6-B and C6-H).

Note : The switch is a normally closed switch.

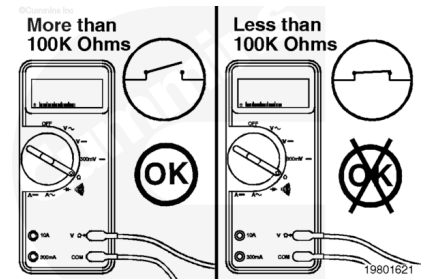
Select the resistance function on the multimeter. Touch the multimeter leads to the terminals on the switch. Toggle the switch to the ON position.



Measure the resistance.

The multimeter **must** show more than 100k ohms, which is an open circuit. If the circuit is **not** open, then there is a short circuit within the switch.

Replace the switch. Refer to the OEM troubleshooting and repair manual for the procedure.

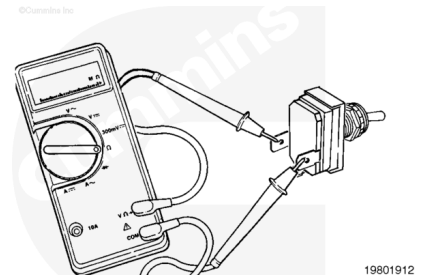


Toggle the switch to the OFF position.

Measure the resistance.

The multimeter **must** show less than 10 ohms, which is a closed circuit. If the circuit is **not** closed, then there is an open circuit within the switch.

Replace the switch. Refer to the OEM troubleshooting and repair manual for the procedure.

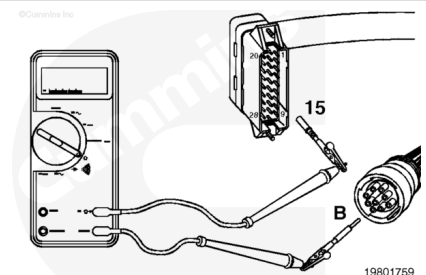


Check the main engine harness portion of the alternate torque/alternate low-idle switch circuit.

Disconnect the ECM connector and the C5 connector.

Select the resistance function on the multimeter.

Touch one of the multimeter leads to pin 15 of the main engine harness connector. Touch the other multimeter lead to pin B of the main engine harness side of the C5 connector.



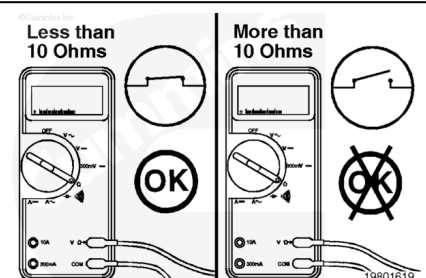
Measure the resistance.

The multimeter **must** show less than 10 ohms, which is a closed circuit. If the circuit is **not** closed, then there is an open circuit in the signal wire.

Repair the main engine harness, or, if necessary, replace it.

Refer to Procedure 019-228

([/qs3/pubsys2/xml/en/procedures/98/98-019-228.html](#)) or 019-043 ([/qs3/pubsys2/xml/en/procedures/98/98-019-043.html](#)).

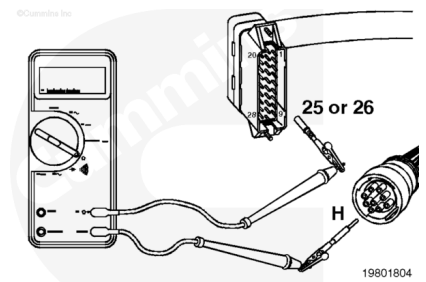


Repeat the above short circuit from pin to pin check for the return wire.

Touch one of the multimeter leads to pin 25 or 26 of the main engine harness connector. Touch the other multimeter lead to pin H of the main engine harness side of the C5 connector.

Measure the resistance.

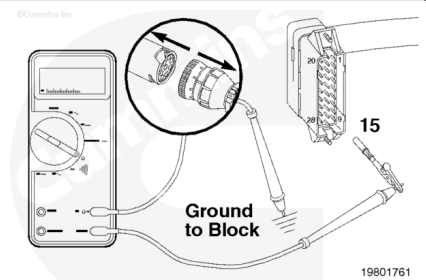
The multimeter **must** show less than 10 ohms.



Check for Short Circuit to Ground

Make sure the ECM connector and the C5 connector are disconnected.

Touch one of the multimeter leads to pin 15 of the main engine harness connector. Touch the other multimeter to a good, clean surface of the engine block.



Measure the resistance.

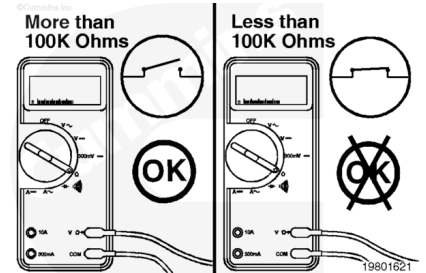
The multimeter **must** show more than 100k ohms, which is an open circuit. If the circuit is **not** open, then there is a short circuit between the wire connected to pin 15 and chassis ground.

Repair the main engine harness, or, if necessary, replace it.

Refer to Procedure 019-228

(/qs3/pubsys2/xml/en/procedures/98/98-019-228.html) or

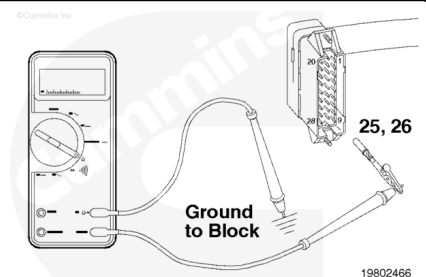
019-043 (/qs3/pubsys2/xml/en/procedures/98/98-019-043.html).



Repeat the above short circuit to ground check for the return wire.

Touch one of the multimeter leads to pin 25 of the main engine harness connector. Touch the other multimeter lead to a good, clean surface of the engine block. Measure the resistance. The multimeter **must** show more than 100k ohm.

Touch one of the multimeter leads to pin 26 of the main engine harness connector. Touch the other multimeter lead to a good, clean surface of the engine block. Measure the resistance. The multimeter **must** show more than 100k ohm.

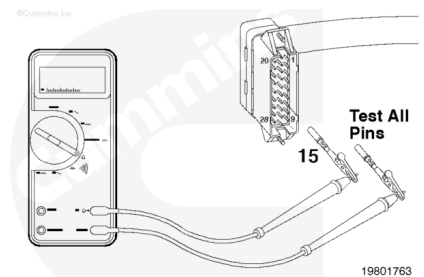


Check for Short Circuit from Pin to Pin

Make sure the ECM connector and the C5 connector are disconnected.

Check for a short circuit between pin 15 of the main engine harness connector and **all** other pins in the connector.

Touch one of the multimeter leads to pin 15 of the main engine harness connector. Touch the other multimeter lead to **all** other pins in the connector, one at a time.



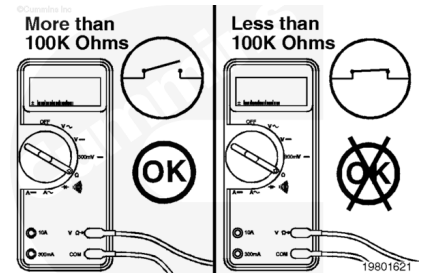
Measure the resistance.

The multimeter **must** show more than 100k ohms, which is an open circuit. If the circuit is **not** open, then there is a short between the wires connected to pin 15 in the main engine harness connector and **any** other pin in the main engine harness connector that measured less than 100k ohms.

Repair the main engine harness, or, if necessary, replace it.

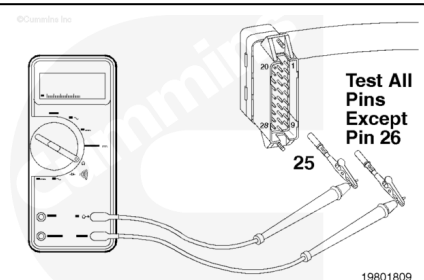
Refer to Procedure 019-228

(/qs3/pubsys2/xml/en/procedures/98/98-019-228.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/98/98-019-043.html).



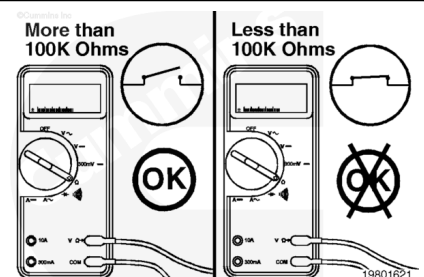
Repeat the above short circuit from pin to pin check for the return wire.

Touch one of the multimeter leads to pin 25 of the main engine harness connector. Touch the other multimeter lead to **all** other pins in the connector, **except** pin 26, one at a time.



Measure the resistance.

The multimeter **must** show more than 100k ohms, which is an open circuit. If the circuit is **not** open, then there is a short between the wires connected to pin 26 in the main engine harness connector and **any** other pin in the main engine harness connector that measured less than 100k ohms.



Repair the main engine harness, or, if necessary, replace it.

Refer to Procedure 019-228

(/qs3/pubsys2/xml/en/procedures/98/98-019-228.html) or
019-043 (/qs3/pubsys2/xml/en/procedures/98/98-019-
043.html).

Repeat the short circuit from pin to pin check for the second
return wire.

Touch one of the multimeter leads to pin 26 of the main engine
harness connector. Touch the other multimeter lead to **all**
other pins in the connector, **except** pin 25, one at a time.

Measure the resistance. The multimeter **must** show more than
100k ohms.

